

Intra-articular knee

NOVA-M1 + NOVA-E1 • physician reference protocol

PER KNEE**2 cc total**

M1 1 cc + E1 1 cc

BILATERAL**4 cc total**

2 cc per joint

NEEDLE**22-25 G**

1.5 in, longer if indicated

OBSERVATION**10-15 min**

post-treatment

A potential approach for intra-articular administration in patients presenting with symptomatic knee degeneration, osteoarthritis, chronic knee pain, meniscal degeneration, or cartilage wear.

Indications

- Mild-to-moderate osteoarthritis
- Chronic knee pain
- Degenerative meniscal pathology
- Cartilage wear without complete joint collapse

Contraindications

- Active infection or septic arthritis
- Uncontrolled coagulopathy
- Known product hypersensitivity
- Intra-articular corticosteroid within four weeks

Procedure

PRE-PROCEDURE

1. Review patient history and available imaging.
2. Obtain informed consent.
3. Verify product preparation and integrity.
4. Position supine, knee flexed ~90°.
5. Prepare skin with chlorhexidine or povidone-iodine.

POST-PROCEDURE — PATIENT INSTRUCTIONS

- Avoid NSAIDs 48-72 h when clinically appropriate
- Avoid alcohol 48-72 h
- Limit strenuous lower-extremity activity 2-3 days
- Cold therapy as needed; resume rehab as tolerated

INJECTION

1. Aspirate effusion if present.
2. Confirm intra-articular placement.
3. Slowly administer 1 cc NOVA-M1 and 1 cc NOVA-E1.
4. Apply sterile dressing after needle withdrawal.

Preferred: anterolateral or anteromedial approach. Alternative: superolateral with knee in extension.

FOLLOW-UP & OUTCOME MEASURES

Clinical assessment at **4-6 weeks**. Consider repeat treatment or adjunctive therapy at **8-12 weeks** based on provider assessment and response.

VAS PAIN

WOMAC

KOOS

FUNCTIONAL STATUS

ACTIVITY LEVEL

REGULATORY

CelluNOVA products are supplied for use by licensed healthcare professionals. CelluNOVA does not direct, prescribe, or control the practice of medicine, and nothing here guarantees a clinical outcome.

CHART DOCUMENTATION · EVALUATION VISIT

PATIENT NAME	DATE OF BIRTH	PATIENT ID / MRN	DATE OF VISIT
EVALUATING PHYSICIAN	INITIAL · FOLLOW-UP VISIT TYPE — CIRCLE	REFERRING PROVIDER, IF ANY	TIME IN / OUT

S SUBJECTIVE

Knee pain —
 CHIEF COMPLAINT

ONSET / DURATION

SEVERITY — VAS 0-10

QUALITY / RADIATION

AGGRAVATING FACTORS

RELIEVING FACTORS

PT · NSAIDs · corticosteroid inj · hyaluronic acid · bracing · arthroscopy
 PRIOR TREATMENT — CIRCLE ALL TRIED · ADD DATES

stairs · kneeling · walking distance · sleep · work · sport
 FUNCTIONAL IMPACT

fever · chills · weight loss · neuro changes — note positives
 PERTINENT ROS

MEDICATIONS — INCL. ANTICOAGULANTS, IMMUNOSUPPRESSANTS

ALLERGIES · RELEVANT PMH — CANCER, INFECTION, PREGNANCY

O OBJECTIVE

BP	HR	TEMP	WEIGHT	GENERAL APPEARANCE
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Inspection — effusion, alignment, quadriceps bulk

Palpation — joint line, patellofemoral tenderness

ROM — flexion / extension, degrees

Stability — Lachman, varus/valgus, McMurray

Gait and squat tolerance

IMAGING / STUDIES REVIEWED — MODALITY, DATE, FINDINGS	BASELINE — VAS /10	BASELINE — WOMAC	BASELINE — KOOS
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A ASSESSMENT

Symptomatic knee OA · chronic knee pain · degenerative meniscus
 WORKING DIAGNOSIS — CONFIRM OR AMEND

ADDITIONAL DIAGNOSES / CODES

☐ Appropriate candidate — indications met, contraindications absent

☐ Alternatives discussed, including conservative care and doing nothing

☐ Non-FDA-approved status explained; no outcome promised

☐ Patient questions answered; understanding confirmed

P PLAN

Proposed treatment. Intra-articular injection, affected knee(s): NOVA-M1 1 cc + NOVA-E1 1 cc per knee, 22-25 G, landmark or ultrasound guidance. Observation 10-15 min. NSAID and alcohol hold explained.

☐ Informed consent (CN-PC) reviewed and signed

☐ Procedure performed today — see CN-OP-KNEE · or scheduled (date right)

☐ Fees, non-coverage by insurance, and refund policy reviewed

☐ Follow-up: assessment at 4-6 wk; repeat consideration 8-12 wk

PHYSICIAN SIGNATURE	PRINTED NAME & LICENSE NUMBER	PROCEDURE DATE, IF SCHEDULED
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PATIENT NAME

DATE OF BIRTH

PATIENT ID / MRN

DATE OF PROCEDURE

PERFORMING PHYSICIAN

Symptomatic knee OA · chronic knee pain · degener...

PRE-PROCEDURE DIAGNOSIS — CONFIRM OR AMEND

START TIME

ASSISTING STAFF

PRE-PROCEDURE CHECKLIST

Check each box before the first draw

☐ Informed consent (CN-PC) signed and on file; SOAP note (CN-SN-KNEE) complete

☐ Screening reviewed — no contraindications present

☐ Time-out performed — correct patient, site, product, route

☐ Five checks read aloud with second staff: ID · product & lot · expiry · volume & route · consent

☐ Baseline outcome measures recorded: VAS pain · WOMAC · KOOS

☐ Site marked and prepped — chlorhexidine / povidone-iodine (circle); chain-of-custody log initiated

BP · HR · TEMP

agent · concentration · volume — write NONE i...

BP · HR · TEMP

PRE-PROCEDURE VITALS

LOCAL ANESTHETIC

POST-OBSERVATION VITALS

PRODUCTS ADMINISTERED

Per knee; double volumes for bilateral treatment

PRODUCT & TARGET	PLANNED	LOT	EXPIRY	GIVEN
NOVA-M1 Intra-articular, affected knee	1 cc / knee			
NOVA-E1 Intra-articular, affected knee	1 cc / knee			

TECHNIQUE

Position: supine, knee flexed ~90° (extension for superolateral)

- Approach: anterolateral or anteromedial; alternative superolateral in extension.
- Aspirate effusion if present; confirm intra-articular placement before injecting.
- Administer M1 then E1 slowly; sterile dressing after withdrawal.

LEFT · RIGHT · BILATERAL — circle

SITE / SIDE

LANDMARK · ULTRASOUND — circle

GUIDANCE

22-25 G, 1.5 in

NEEDLE USED

target 10-15 min

OBSERVATION — MINUTES

NARRATIVE — FINDINGS, ASPIRATE, DEVIATIONS FROM PROTOCOL, PATIENT RESPONSE

POST-PROCEDURE & DISPOSITION

Tolerated: WELL · WITH EVENTS — circle

Discharged: STABLE · AMBULATORY · WITH ESCORT — circle

☐ Patient note CN-PN-KNEE reviewed and given at discharge

COMPLICATIONS / ADVERSE EVENTS — WRITE "NONE" IF NONE

Any adverse event also requires the adverse-event response steps in this book.

Follow-up plan. Clinical assessment at 4-6 weeks; consider repeat treatment or adjunctive therapy at 8-12 weeks based on response.

FOLLOW-UP SCHEDULED FOR

PHYSICIAN SIGNATURE

PRINTED NAME & LICENSE NUMBER

DATE & END TIME

Today you received an injection of cell and exosome products into your knee joint. The products used are made from donated birth tissue and are not FDA-approved treatments. This page tells you what is normal in the days ahead, how to care for yourself, and when to call us.

DATE OF TREATMENT

SITE(S) TREATED

NEXT VISIT

WHAT TO EXPECT

Changes are gradual — do not judge the result in the first days

FIRST 72 HOURS

Soreness, warmth, or mild swelling in the knee is common and usually settles in two to three days. A short flare of your usual pain can happen.

WEEKS 1-4

Ease back into normal activity as comfort allows and restart physical therapy or home exercises as advised.

WEEKS 4-12

This is when progress is assessed. Keep your follow-up visit so we can measure how the knee is responding.

DO

- Take it easy the rest of today; gentle walking is fine
- Use cold packs 10-15 minutes at a time for comfort
- Take acetaminophen (Tylenol) for soreness if needed
- Resume physical therapy or home exercises as advised

AVOID

- Anti-inflammatory pain relievers (ibuprofen, naproxen, aspirin for pain) for 72 hours
- Alcohol for 72 hours
- Squatting, impact, or strenuous lower-body exercise for 2-3 days
- Hot tubs or soaking the injection site for 24 hours

CALL US RIGHT AWAY IF

For an emergency, call 911 first

- The knee becomes increasingly hot, red, and swollen rather than better
- Fever above 101 °F or shaking chills
- Calf pain or swelling, or shortness of breath
- Severe pain not controlled by acetaminophen and cold packs

CLINIC PHONE

AFTER-HOURS PHONE

NOTES FROM YOUR VISIT